



Cognitive digital twin-enabled multi-robot collaborative manufacturing: Framework and approaches

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ABSTRACT

Multi-robot collaborative manufacturing (MRCMfg) is vital to ensure effective and efficient operation in the manufacturing industry. In recent years, enabled by the new generation of information technology, represented by the digital twin (DT), the digitalization of MRCMfg has progressed steadily. However, more complex manufacturing tasks and more customized manufacturing demands pose challenges to MRCMfg, which limit the applicability of MRCMfg in practice. Recent successes in intelligent manufacturing show that empowering DT with the intelligence and cognition capabilities to adapt to the above challenges is a promising approach. Inspired by that, a cognitive digital twin (CDT) framework with multiple artificial intelligence algorithms for MRCMfg is proposed. The framework consists of three critical spaces: (i) Physical-virtual space: multi-agent system-based DT; (ii) Cognition space: multi-behavior tree (BT) cognition models; (iii) Data space: data-driven cognition evolution. In the data space, multimodal perception is a important way to improve the cognition capabilities of MRCMfg. To alleviate the multimodal data quality requirements as well as to improve the robustness of the perception model, an incomplete multimodal Transformer with deep autoencoder (IMTDAE) approach is proposed. The effectiveness and feasibility of this framework and IMTDAE are verified by case studies. The results provide compelling evidence that this framework has the potential to significantly enhance the intelligence and cognition capabilities of MRCMfg.

1. Introduction

In recent years, the manufacturing industry has witnessed a transformative shift with the integration of advanced robotics. These innovations have significantly enhanced productivity, enabling companies to optimize operations, reduce errors, and expedite production processes (Karnik et al., 2022; Wang et al., 2019). Robotic manufacturing (RMfg) in manufacturing not only streamlines tasks that are labor-intensive but also opens up numerous new avenues for human employment. By handling repetitive and hazardous tasks, RMfg frees up human workers to engage in more complex and creative roles, driving innovation and growth within the industry. However, due to the increasing complexity of manufacturing tasks and the growing demand for customized products, it is not rare for RMfg to struggle with handling variations in the manufacturing processes (Thomas et al., 2018). With the recent upsurge in robotics and artificial intelligence, multi-robot collaborative manufacturing (MRCMfg) has attracted extensive interest from researchers (Karnik et al., 2022; Pérez et al., 2020). Compared with RMfg,

MRCMfg represents a sophisticated, flexible, and collaborative manufacturing approach that leverages the collective capabilities of multiple robots to achieve unparalleled efficiency and productivity.

Early developed approaches are primarily focused on the digitalization of RMfg. Among them, Digital twin (DT) plays a pivotal role in the development of Industry 4.0 (Tao et al., 2019). DT is a comprehensive lifecycle simulation process that integrates multiple disciplines, scales, physical quantities, and probabilities (Shafte et al., 2012). It monitors changes in RMfg, simulates its behavior, assesses its state, predicts trends, optimizes performance, and controls operations through multi-dimensional virtual models and real-time virtual-reality interactions (Tao et al., 2021). Meanwhile, this leaves a significant technological divide between digitizing MRCMfg and realizing its intelligence and cognition. RMfg systems without intelligence and cognition capabilities rely exclusively on human experience to deal with the various complexities that occur during the manufacturing process. Since the intelligence and cognitive transformation of MRCMfg is a complex problem involving data interaction, environment perception, knowledge transfer,

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and autonomous decision-making, research in this area is still limited. How to manage multi-dimensional virtual models, construct cognition models, perform knowledge transfer, make autonomous decisions, realize seamless collaborative operations, and design an intelligent and cognitive framework for MRCMfg has become an extremely challenging problem.

Multimodal data is the basis for realizing accurate perception and autonomous decision-making. However, the collection of multimodal data is challenging. Meanwhile, the robustness of perception models obtained from multimodal data is extremely poor. Signal-to-noise ratio and synchronization of multimodal data are difficult to ensure in complex manufacturing environments. While using multimodal data for accurate perception, it is also worthwhile to pay attention to how to improve the robustness of perception models in these complicated environments.

To address the above issues, the topic in this paper focuses on a cognitive digital twin (CDT) framework for MRCMfg that integrates DT, knowledge-driven approaches, and data-driven approaches. CDT, in contrast to traditional DT, has the unique capability to seamlessly integrate heterogeneous manufacturing elements within a manufacturing system. These digital models not only replicate the physical aspects of these elements but also enhance them through cognitive approaches. These cognition capabilities enable CDT to actively support and engage in autonomous activities within the manufacturing process (Lu et al., 2022). These capabilities are achieved through a unified description form of multidimensional cognition models and cognition evolution driven by data and knowledge collaboration (Zheng et al., 2022). This cutting-edge concept holds tremendous potential in revolutionizing various industries and enhancing the capabilities of MRCMfg. Meanwhile, Transformer has been successfully applied to computer vision (CV) as well as natural language processing (NLP) (Dosovitskiy et al., 2021; Vaswani et al., 2017). To improve the perception accuracy in complex environments, an incomplete multimodal Transformer with deep autoencoder (IMTDAE) model is proposed, which combines data modal random masking and deep autoencoder (DAE)-based data repair. IMTDAE is validated by multimodal perception for screw-tightening, proving its excellent perception performance and robustness.

The steps of the CDT framework for MRCMfg are detailed below. Firstly, the virtual models of MRCMfg are constructed by leveraging the power of DT. Then, the robot agents, task agents, and workstation agents are constructed by the multi-agent system (MAS). Following that, multidimensional cognition models based on the Behavior Tree (BT) are designed and constructed to facilitate the interaction and collaboration between various agents in MRCMfg (Hu et al., 2021). BT, as a knowledge-based engineering technique, is a structured, ordered directed acyclic graph with a structure similar to the decision tree (Colledanchise & Ögren, 2018). BT effectively creates modular, reactive complex systems, driving advancements in computer game programming, AI, and robotics (Colledanchise & Ögren, 2018). Finally, enabled by data-driven algorithms, including deep learning (DL), transfer learning (TL), and reinforcement learning (RL), the cognition models are evolved to enhance the intelligence and cognition capabilities of MRCMfg (Mnih et al., 2015; Pan & Yang, 2010; Vaswani et al., 2017). The case studies with different goals and complexities evidence the superiority of the CDT framework. They also exemplify the industrial applicability of the CDT framework in real-world environments.

The original and main contributions of this paper are summarized as follows:

- A CDT framework for MRCMfg is proposed. Virtual models in DT and knowledge/data-driven cognition models are leveraged in this framework to boost the overall performance of MRCMfg in terms of multimodal perception, knowledge transfer, and autonomous decision-making.

- A unified description form of cognition models based on BT is introduced to accommodate the distinctive characteristics of various manufacturing elements within the MRCMfg and facilitate the seamless integration of data-driven approaches and knowledge-driven approaches.
- The cognition evolution of MRCMfg based on multiple data-driven approaches is introduced to enable MRCMfg to adapt to the dynamic manufacturing environment. Among them, an IMTDAE approach is proposed to realize incomplete multimodal perception.
- The proposed cognition evolution approaches are validated by four case studies with different goals to demonstrate the effectiveness and feasibility of the proposed CDT framework.

The rest of this paper is organized as follows. In Section 2, related works are reviewed. In Section 3, the CDT framework for MRCMfg is explained in detail. In Section 4, the process of cognition model construction and its cognition evolution are presented. In Section 5, case studies and analyses are elaborated. In Section 6, future works are discussed and conclusions are drawn.

2. Related work

In this section, related works about RMfg are reviewed from the following two aspects: DT in RMfg and cognitive approaches for RMfg.

2.1. DT in RMfg

Before the emergence of the DT, simulation technology represented by cyber-physical systems (CPS) was the dominant approach for digitizing RMfg. CPS integrates computing, communication, and control to enable reliable, secure, and efficient real-time monitoring of physical entities (Khalid et al., 2018; Tao et al., 2019). Compared to DT, CPS requires higher levels of system responsiveness and real-time adjustment capabilities. However, these technological demands also result in increased operational costs and resource consumption, which can make practical application in production processes challenging. However, DT has revolutionized the simulation of RMfg, taking it to a new stage of high-fidelity virtual model construction and real-time virtual-real interaction of data. The emergence of DT has significantly enhanced the capabilities and effectiveness of simulation (Baratta et al., 2024).

From the perspective of physical entity structure, DT can manage the lifecycle in three dimensions: component, product, and system (Boschert & Rosen, 2016; Lu et al., 2022). The component DT focuses on individual parts and their performance, allowing for precise monitoring, optimization, and maintenance. To efficiently identify abnormal states in the manufacturing system, the component DT provides a novel method for training a classifier by simulating faults within the power electronic converters (Wileman et al., 2023). This approach is advantageous because it is challenging to replicate such faults in a physical device. In robotics, component DT is utilized to generate virtual data in fault states by constructing local virtual models of robot joints. Diagnostic classifiers of joint states, based on this virtual data, can ensure accurate execution, stable operation, and accident prevention for the robot (Song et al., 2023). Additionally, component DT can use low-cost local sensors and virtual models to detect positioning errors in robot joints and reduce these errors by adjusting the joint angles (Wu et al., 2022).

The product DT encompasses an entire physical product, such as a robot, providing insights into its behaviors and interactions within different environments. It enables comprehensive testing, design validation, and performance optimization before actual production or deployment. Product DT is increasingly applied to the simulation and optimization of specific manufacturing tasks such as motion planning, and robot grasping. Liu et al. (2022) created an assembly-oriented robot grasping virtual environment and proposed a DT-enabled approach for transferring grasping skills to the physical robot. Lv et al. (2021)

proposed a DT-based framework of human-robot collaborative manufacturing (HRCMfg), which obtains the action sequence in a dynamic environment by integrating all kinds of data from virtual space and evaluating the quality of manufacturing. For HRCMfg, to ensure human safety, Choi et al. (2022) proposed a minimum safe distance safety-aware approach based on DT and mixed reality (MR). To improve the accuracy of the moving trajectory of the robot, Liu et al. (2022) presented a trajectory optimization approach based on DT, which trains the robot in the virtual environment and applies it to the physical space.

The system DT integrates multiple robots and complex manufacturing environments, offering a holistic view of complex systems and enabling advanced analysis and decision-making. It can capture dynamic interactions and dependencies among different elements within a system, and offer insights into overall system performance, efficiency improvements, and predictive capabilities. Zhu et al. (2022) proposed a dynamic reconfiguration optimization approach for RMfg based on DT, which promotes reconfiguration accuracy and production efficiency. Furthermore, Fan et al. (2021) presented a general DT visualization architecture for flexible RMfg. The architecture can effectively realize the design, development, debugging, visual monitoring, maintenance, as well as remote assistance of RMfg. To meet the new demands of production, Zhang et al. (2021) presented a five-dimensional fusion model of a DT virtual entity for RMfg to support automatic reconfiguration. By combining DT with digital technologies such as augmented reality (AR) and virtual reality (VR), more intuitive and flexible RMfg can be achieved. Pérez et al. (2020) presented a novel approach for the design, implementation, and monitoring of RMfg based on DT and VR, which can train operators and monitor the state of RMfg. To minimize the duplication of learning efforts among operators from various manufacturing sectors and to provide intuitive control of robots, Li et al. (2022) utilized AR and DT to create a virtual model of the RMfg.

This paper aims to highlight the unique capabilities of system-level DT in complex MRCMfg, and the significant impact they have on the intelligence and cognition capabilities of MRCMfg when combined with knowledge and data-driven approaches.

2.2. Cognitive approaches for RMfg

Early works on cognitive approaches for RMfg were mainly based on data-driven approaches and knowledge-driven approaches.

In this paper, knowledge-driven approaches refer to various structured semantic knowledge representation methods, including knowledge graphs (KG), ontology, BT, and finite state machines. These approaches can describe the complex interrelationships among multiple devices in an MRCMfg system and guide the system's efficient operation by integrating multidimensional knowledge. These knowledge-driven approaches have been widely used in expert systems and recommendation systems. In contrast to the data-driven approach, the knowledge-driven approach offers superior interpretability and applicability.

In the manufacturing field, KG systematically and semantically organizes manufacturing knowledge from multi-source heterogeneous data in a graph manner. Based on such characteristics of KG, numerous manufacturing tasks significantly improve their performance (Zheng et al., 2021). To improve the swift reaction of RMfg to uncertain market demands, Li & Wang, 2022 proposed an approach to add an intelligent dimension to DT by enabling learning capability for RMfg using KG. To achieve intelligent resource optimization allocation under dynamic customized orders demand, Zhou et al. (2021) presented a unified KG-driven resource allocation optimization method to fully integrate the complex data of manufacturing resources. To integrate and fuse multi-dimensional and heterogeneous data in a manufacturing scenario, Liu et al. (2022) proposed a KG-based data representation approach for RMfg to enable intelligent manufacturing decision-making from a global perspective. In addition to KG, ontology, BT and other knowledge-driven approaches have also received extensive attention. To enable robots to operate in unpredictable manufacturing scenarios, Styruud, Iovino,

Norrlöf, Björkman, & Smith, 2022 proposed an approach for creating BTs by combining automated planning and machine learning in robotic assembly manufacturing tasks. Hu et al. (2021) empowered Automated Guided Vehicles (AGVs) with adaptive strategy selection capability in collision cases by constructing BTs of AGVs and traffic. Zheng, An, et al. (2023) proposed a knowledge-based engineering approach to enable users to customize RMfg as well as alleviate the burden on engineers. To improve the efficiency of robot programming and manufacturing quality, Zheng et al. (2022) proposed a knowledge-based program-generation approach for RMfg. Bernardo et al. (2023) presented a novel motion planning approach based on the Semantic Web Rule Language (SWRL) for reasoning the ontological knowledge in complex manufacturing tasks.

With the more successful application of artificial intelligence in various manufacturing scenarios, data-driven approaches for improving intelligence and cognition capabilities have been actively explored. In RMfg scenarios, the most commonly used data-driven methods include DL, TL, and RL. To design suitable RMfg systems for enterprises, Zheng, Du, et al. (2023) proposed a common data model, which enables different agents to communicate, interact, and negotiate with each other according to their experiences and knowledge. Ying et al. (2021) proposed a bi-directional rapidly exploring random tree integrated with the long short-term memory method to plan collision-free trajectories of dual-arm assembly robots in complex operational environments. Droniou et al. (2015) proposed an unsupervised DL approach to enable RMfg to learn autonomously in changing manufacturing environments. Roza et al. (2016) proposed a framework for teaching robots collaborative skills and cognition capability from demonstrations, which can make robots carry out a variety of manufacturing tasks with operators. Lv et al. (2022) proposed a framework of HRCMfg based on TL to realize rapid design and verification of product assembly strategy, which consists of strategy generation, similarity evaluation, and strategy transfer. Zhang et al. (2022) presented a human-robot collaborative RL approach to optimize the human-robot subtask allocation sequence. Yao et al. (2023) presented a fatigue evaluation network for HRCMfg to generate a task reallocation strategy based on RL.

2.3. Research gap

With the development of DT, the complexity of virtual sub-models incorporated within DT has been increasing. These sub-models often involve intricate structures and diverse functionalities. Therefore, designing a unified description form to integrate, fuse, and manage virtual sub-models within DT is indeed a research hotspot. In addition, a significant gap persists between the digitization and the intelligence of MRCMfg. Neither knowledge-driven nor data-driven approaches can significantly improve the intelligence and cognition capabilities of MRCMfg. Knowledge-driven approaches are often limited by the difficulty of autonomously evolving to adapt to new manufacturing environments. Data-driven approaches are often limited by the lack of sufficient data and the difficulty of data acquisition. Therefore, it is necessary to combine knowledge-driven approaches and data-driven approaches to address the above challenges. The concept of CDT provides new ideas to address these challenges (D'Amico et al., 2022; Mortlock et al., 2022). There are still four problems that need to be addressed: (i) designing a CDT framework; (ii) constructing unified structural cognition models for multiple core elements; (iii) presenting the cognition evolution approaches for cognition models; (iv) improving the robustness of data-driven multimodal perception methods.

3. The CDT framework for MRCMfg

We design a CDT framework for MRCMfg, which is shown in Fig. 1. The CDT framework encompasses three interrelated spaces, namely the physical-virtual space (MAS-based DT), the cognition space (multi-BT cognition models), and the data space (data-driven cognition evolution).

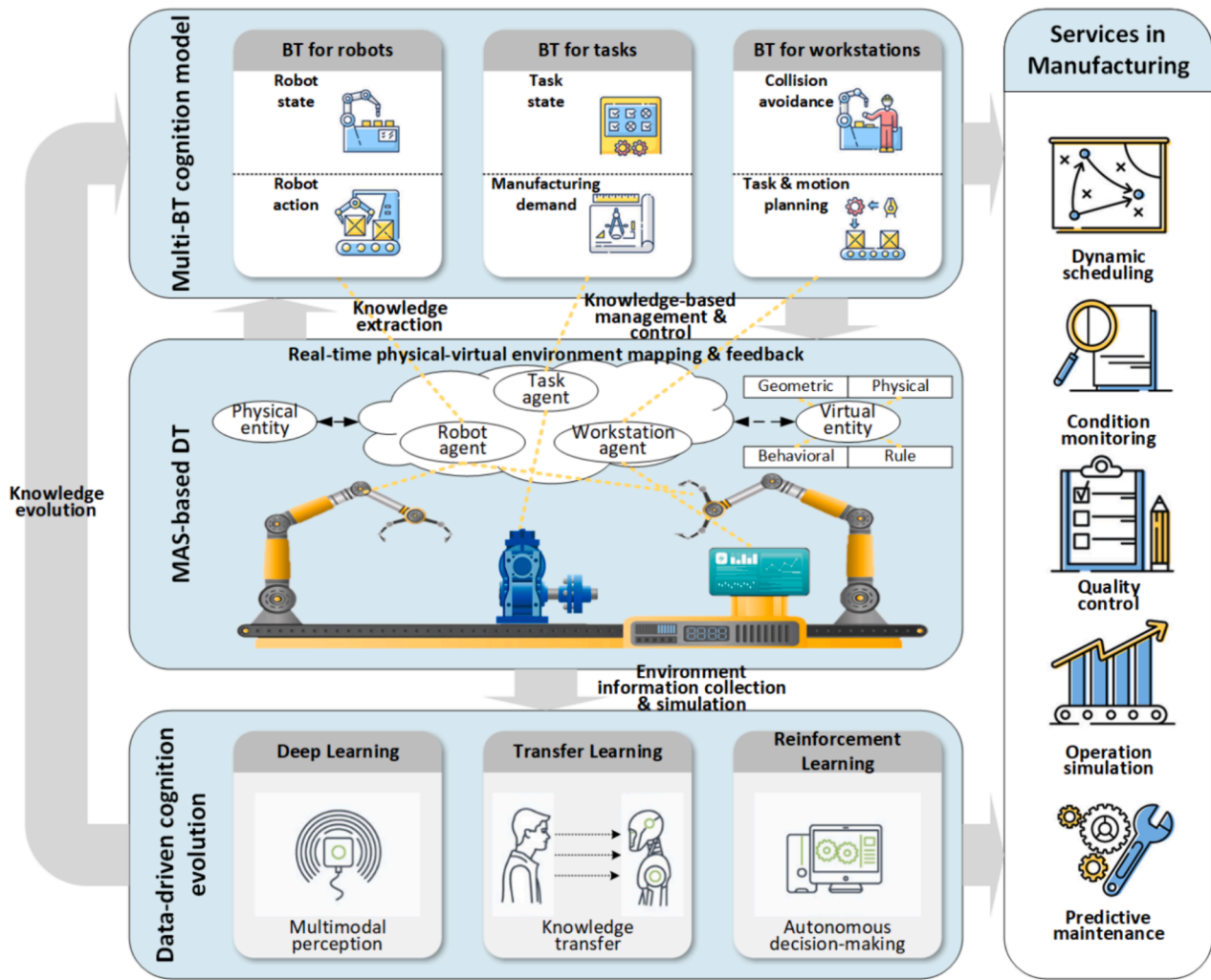


Fig. 1. The CDT framework for MRCMfg.

By integrating and fusing the three key spaces within the framework, the MRCMfg is empowered to deliver a range of sophisticated services within complex manufacturing environments.

3.1. Physical-virtual space: MAS-based DT

The MAS-based DT in physical-virtual space is the basis of the whole CDT framework. In this space, by constructing and integrating geometric models, physical models, behavioral models, and rule models of multiple manufacturing elements in the MRCMfg, the virtual entities and virtual environment of the MRCMfg are created. Based on advanced sensing and communication technologies, the DT of the MRCMfg can realize bidirectional dynamic interaction between physical entities and virtual entities, high-fidelity mapping, and dynamic planning (Liu et al., 2021). Then, to empower manufacturing elements in the MRCMfg with manufacturing capacities of real-time collaboration, autonomous planning, and autonomous decision-making, the MAS technology is employed to create specialized agents for the key manufacturing elements in the DT of the MRCMfg. There are three key manufacturing element agents utilized in the MRCMfg: robot agents, task agents, and workstation agents.

3.2. Cognition space: multi-BT cognition model

BT is utilized to construct cognition models for three agents in MRCMfg. However, due to the different roles of robot agents, task

agents, and workstation agents, their operational characteristics need to be analyzed separately. Then, tailored cognition models can be constructed using BT for each agent. The cognition model based on BT for robot agents consists of the states and actions of the robots. It can represent all working conditions that robots may encounter in the manufacturing process by combining different states and actions (Hu et al., 2021). For task agents, the cognition model based on BT comprises the manufacturing task states and manufacturing demands. With this cognition model, the task agents can rationally decompose manufacturing tasks into different subtasks. For workstation agents, the cognition model based on BT consists of two parts: collision avoidance and task & motion planning. This cognition model enables workstation agents to select suitable subtask execution strategies.

3.3. Data space: data-driven cognition evolution

To enable cognition models to evolve autonomously and adaptively, various data-driven approaches are utilized. These approaches evolve cognition models in the MRCMfg from three perspectives, including multimodal perception based on DL, knowledge transfer based on TL, and autonomous decision-making based on RL. DL can be utilized to process multimodal data from various sensors and understand the state of the manufacturing environment (Liu et al., 2019). Furthermore, data deficiency often poses a challenge for MRCMfg to rapidly adapt to new manufacturing tasks (Cao et al., 2021). The knowledge transfer based on TL can effectively apply manufacturing knowledge from previous

experiences, and virtual environments to MRCMfg. Additionally, as manufacturing tasks become more complex and customized, it becomes challenging for MRCMfg to plan manufacturing task sequences. Thus, using RL to improve the autonomous decision-making capability of MRCMfg is crucial.

4. The developed methodology

4.1. Manufacturing task description

In the following sections of this paper, we will delve into a concrete example that showcases the application of the CDT framework within an MRCMfg system depicted in Fig. 2. There are two distinct robots in the MRCMfg system. The robot on the left is a UR5e robot, equipped with an Onrobot Screwdriver end-effector. This particular end-effector enables the UR5e robot to undertake critical operations, including picking a screw, tightening a screw, and loosening a screw. In contrast, the robot on the right is a UR5 robot, equipped with a Robotiq 2F140 end-effector. This end-effector empowers the UR5 robot to grip and transport various parts in the manufacturing process. Specifically, we will explore how this framework empowers the MRCMfg system with enhanced intelligence and cognition capabilities. This paper takes the manufacturing task depicted in Fig. 3, as an instance. The MRCMfg system is responsible for tightening or loosening 36 screws on the manufacturing platform. Consequently, this manufacturing task can be broken down into 36 subtasks.

The screw-tightening and loosening tasks undeniably constitute a significant portion of automated manufacturing processes. Statistical data underscores their prevalence, indicating these tasks comprise approximately 35 % of typical assembly and disassembly tasks in various products (Jia et al., 2019; Martin-Vega et al., 1995). Table 1 enumerates some completion states associated with screw-tightening and loosening tasks, shedding light on the complexity of these tasks. Considering the unfavorable conditions and uncertainties that may arise, the challenges faced by MRCMfg systems can be divided into three aspects:

- Completion state perception: In the screw-tightening and loosening tasks, the abnormal completion state of the task can lead to adverse consequences, such as a decrease in manufacturing efficiency and damage to the robot.
- Data sample deficiency: In the screw-loosening tasks, the challenge lies in comprehensively collecting data on various completion states

due to the high uncertainty and low automation level associated with such tasks.

- Subtask sequence decision-making: In the screw-loosening tasks, since different subtask completion states have different effects on subsequent robot motions, it is important to plan the subtask sequence in real-time based on the real-time completion states of the subtasks to improve manufacturing efficiency.

4.2. Mas-based DT

The DT modeling for RMfg has been studied extensively (Ramasubramanian et al., 2022). Geometric, physical, behavioral, and rule models of the MRCMfg system are constructed based on previous research (Zhang et al., 2021). Virtual models are constructed by Mujoco and real-time interaction between the virtual and real data is realized through the Robot Operating System (ROS) and URScript (St-Onge & Herath, 2022; Todorov et al., 2012). There are three key manufacturing elements in the MRCMfg system: robots, tasks, and workstations. These three key manufacturing elements have a direct impact on the production efficiency and quality of the MRCMfg system. The interaction mechanism of multiple agents is illustrated in Fig. 4.

Task agents are responsible for decomposing complex manufacturing tasks into smaller, manageable subtasks. They communicate subtasks to the workstation agent. Workstation agents further break down subtasks into a series of specific execution steps based on manufacturing demands, which are then assigned to different robot agents for implementation. Robot agents receive specific execution steps from workstation agents and take on the responsibility of planning and simulating in the virtual environment. After simulating the execution process, robot agents provide feedback to the workstation agents. Based on the feedback, workstation agents optimize relevant strategies for the execution processes. Ultimately, workstation agents take charge of controlling the robot agents to execute the optimized motion, ensuring the completion of the manufacturing tasks. This interaction mechanism allows for effective task decomposition, planning, simulation, optimization, and execution.

4.3. Cognition models construction

To combine the multi-dimensional DT virtual models with MAS, this section proposes a unified construction method of cognition model based on BT.

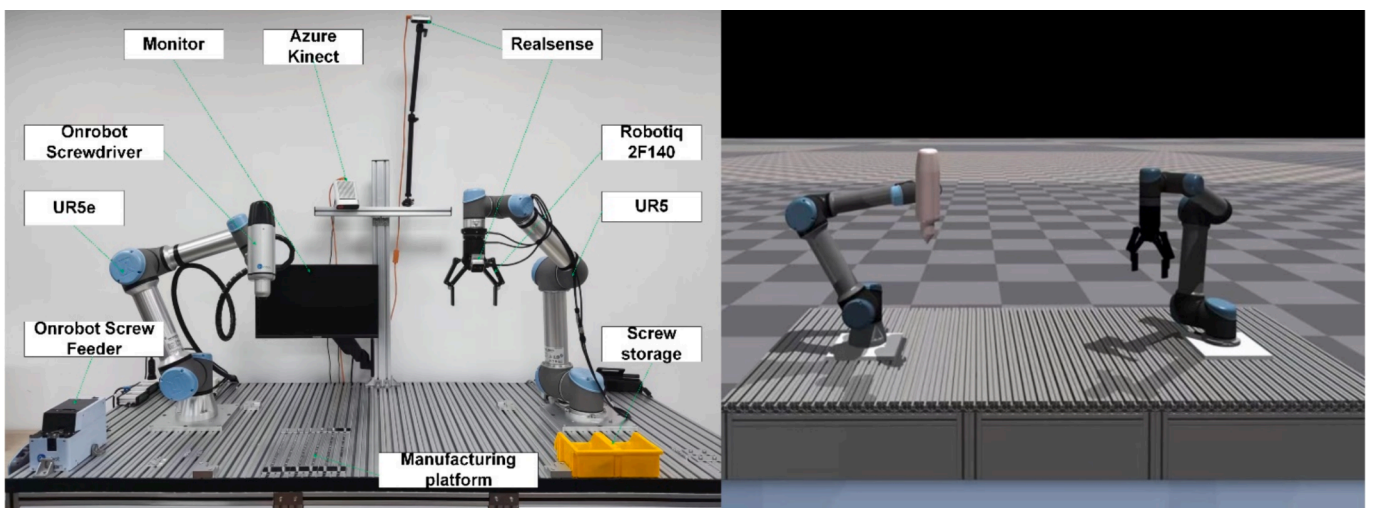


Fig. 2. Physical entity and virtual model of the MRCMfg system.

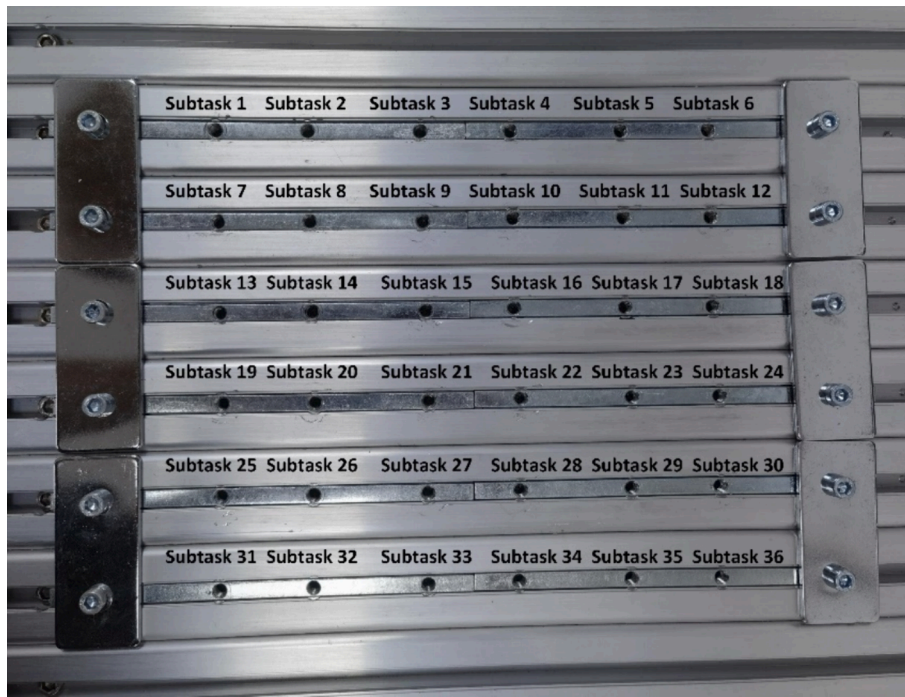


Fig. 3. The manufacturing platform for screw-tightening and loosening.

Table 1
Completion states in screw-tightening and loosening tasks.

States	Types	Description
Normal	Normal	The robot can successfully tighten or loosen screws.
Stripped	Abnormal	The thread of the screw is damaged, and the robot can not tighten or loosen screws.
Damaged	Abnormal	The thread of the metal platform is damaged, and the robot needs to use far more torque than the rated torque to tighten or loosen screws.
Occupied	Abnormal	An extra component is erroneously placed on the metal platform, and the robot can not tighten the screws.
Missing	Abnormal	There is no screw, and the robot can not tighten or loosen screws.

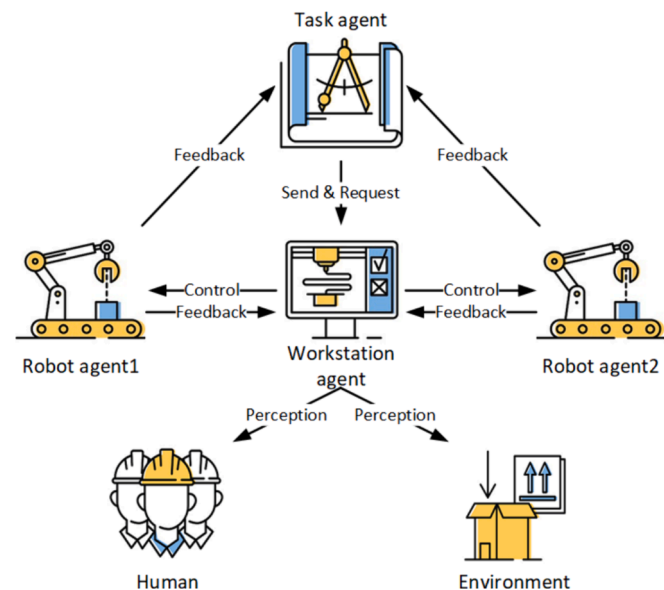


Fig. 4. The interaction mechanism of multiple agents in the MRCMfg system.

4.3.1. Behavior tree

Structurally, BT comprises a root node, multiple intermediate nodes, and multiple leaf nodes. In terms of functionality, intermediate nodes can be categorized into four composite node types (sequence, selector, parallel, and decorator), while leaf nodes are categorized into two node types (state and action). By combining these various node types, BT can effectively represent intricate decision-making processes. In this section, different node types are illustrated as depicted in Table 2. BT processes tick signals to execute nodes, and each node responds with “Running,” “Success,” or “Failure” based on its current state and the outcome of its execution.

- If the node is actively executing, it immediately returns “Running” to the parent node.
- If the node successfully achieves its execution goal, it immediately returns “Success”.

Table 2
Node types in the BT.

Node type	Symbol	Success	Failure	Running
Sequence	Sequence	If all child nodes return success.	If one child node returns failure.	If one child node returns running.
Selector	Selector	If one child node returns success.	If all child nodes return failure.	If one child node returns running.
Parallel	Parallel	If M child nodes return success.	If $N-M$ child nodes return failure.	Else.
Decorator	Decorator	Depending on the function of the decorator.	Depending on the function of the decorator.	Depending on the function of the decorator.
State	State?	If true.	If false.	Never.
Action	Action	Action is completed.	Action can not be completed.	Action is being executed.

- If the node fails to achieve its execution goal, it immediately returns “Failure.”

4.3.2. Robot agents

A structured approach is employed, utilizing the “state-action-state” notation to represent the process of performing these subtasks. The states and actions of the robots within the MRCMfg system are defined in Tables 3 and 4. It is worth noting that the specific options for “STEP-TYPE” and actions are determined by the manufacturing capabilities of the robot. In this section, we build the cognition model for robots that are intuitively designed based on BT. To realize a systematic and hierarchical organization of state and action, the cognition model for robot agents based on BT is developed as illustrated in Fig. 5. Robot agents constantly check the cognition models to confirm the current state and the actions that need to be performed.

4.3.3. Task agents

Cognition models for the task agents encompass two essential components: task states and manufacturing demands. The task state depends on the completion state of each subtask, while the manufacturing demands primarily encompass the manufacturing sequence, manufacturing constraints, and manufacturing quality. Throughout the execution process of tasks, the task agents consistently monitor the subtasks’ completion states. This serves two critical purposes: ensuring that each subtask is accomplished as intended and determining the most appropriate subtask for the next execution subtask. The cognition model for the manufacturing task in Fig. 3. is shown in Fig. 6.

4.3.4. Workstation agents

As shown in Fig. 7, the primary focus of the cognition model for the workstation agents revolves around addressing two pivotal challenges encountered in the MRCMfg system: collision avoidance and task & motion planning. In the realm of collision avoidance, robot agents are equipped with algorithms and sensors to detect potential collisions with human operators and other objects, promptly signaling workstation agents when such risks are identified. Within the domain of multi-robot task & motion planning, workstation agents assume the responsibility of coordinating robot movements. These task & motion planning strategies are tailored to meet specific objectives and may encompass: minimizing energy consumption, minimizing motion time, smoothing the path, or minimizing torque variation.

4.4. Data-driven cognition evolution

This section delves into the methodologies leveraged to improve the intelligence and cognition capabilities of the MRCMfg system. These approaches encompass the following key functions: (i) incomplete multimodal perception based on DL; (ii) knowledge transfer based on TL; and (iii) autonomous decision-making based on RL.

Table 3
The state definition of robot agents.

State	Option	Description
IDLE	True; False	The robot is idle and able to receive a new subtask.
WORKING	True; False	The robot is working and can not receive a new subtask.
COLLISION	True; False	The robot is in a collision state and all operations are disabled.
STEP-TYPE	None; Move; Tighten; Loosen; Grasp; Release	The options of STEP-TYPE are related to the manufacturing capability of the robot.
COMPLETION	Successful; Failed	The completion state of the subtask.
FINISHED	True; False	The completion state of all subtasks.

Table 4
The action definition of robot agents.

Action	Description
Turn to the target state	The state of the robot turns to the target state.
Move to the initial position	The robot plans the trajectory and moves to the initial position.
Move to the target position	The robot plans the trajectory and moves to the target position.
Receive a subtask	The robot receives a subtask.
Check the completion	The robot checks the completion of the subtask.
Pick a screw	The robot picks a screw from the screw feeder.
Tighten a screw	The robot tightens a screw.
Loosen a screw	The robot loosens a screw.
Open the gripper	The robot opens the gripper and grasp the target object.
Close the gripper	The robot closes the gripper and releases the target object.
Stop	The velocity of the robot goes down to 0.

4.4.1. Incomplete multimodal perception based on DL

To enhance the incomplete multimodal perception capability of the MRCMfg system, this subsection focuses on generating and deploying incomplete multimodal perception nodes within the cognition model of the robot agent.

For state perception of MRCMfg, the analysis of multimodal data based on the Transformer plays a pivotal role in unraveling the intricate physical phenomena in the manufacturing industry. Traditionally, torque data analysis has been prioritized as a primary way to perceive and comprehend the screw-tightening process (Jia et al., 2019). However, it is difficult to fully characterize the manufacturing process with a single data modality. The combination of diverse data modalities contributes to a more comprehensive grasp of the underlying physical phenomena in screw-tightening tasks. Since it is difficult to know definitively which data modalities will be most effective, it is common to collect multiple data modalities extensively for multimodal perception. Therefore, this section provides a comprehensive analysis of the screw-tightening tasks by using 8-modality data from the MRCMfg system, as shown in Table 5. Ensuring synchronization of multimodal data collection is challenging in complex manufacturing systems. Anomalies in the collection of any data modality will seriously affect the effectiveness of the multimodal perception model. Therefore, this section proposes an innovative IMTDAE approach to improve the effectiveness of multimodal perception models in the case of incomplete multimodal data.

At the data collection and pre-processing stage, it is worth noting that to construct a unified Transformer model for the screw-tightening and loosening tasks, the data modalities of the screw-tightening and loosening tasks are required to be consistent. Therefore, the multimodal data undergoes padding processing, ensuring uniform size of multimodal data. The size of the 8-modality data is padded to 8×2688 . This step is imperative to maintain the consistency of feature maps within the Transformer model. The architecture of the IMTDAE is elucidated in Fig. 8. The IMTDAE is composed of four primary modules: the incomplete multimodal data processing module, the embedding module, the Transformer encoder module, and the multilayer perceptron (MLP) head module (Dosovitskiy et al., 2021).

The incomplete multimodal data processing module consists of two main components: random data modality masking and data modality correlation matrix generation. This module simulates anomalous collection by randomly masking 0–2 data modalities from each batch in the training dataset. Then, DAE is utilized to generate a correlation index between each data modality and the other data modalities. The DAE is mainly composed of an encoder and a decoder, whose main purpose is to convert the input x into the intermediate variable y , and then convert y into the output x' (Hinton & Salakhutdinov, 2006). DAE makes the input x and output x' similar by optimizing the loss function. In this module, a data modality i^{th} is fed into the DAE. The DAE makes the decoder output similar to another data modality j^{th} by continuously learning the relationship between data modalities in the training

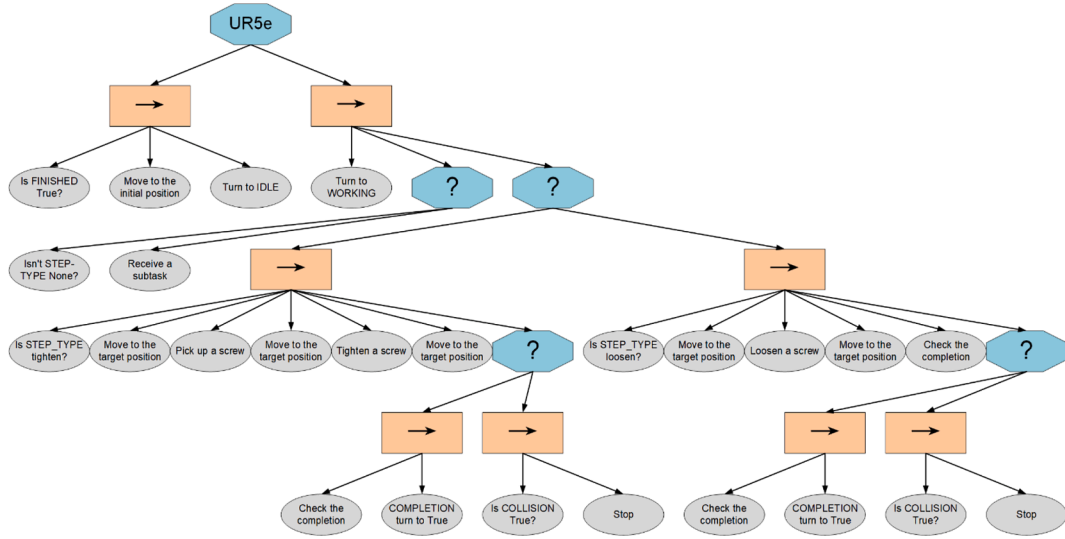


Fig. 5. The cognition model for UR5e based on BT.

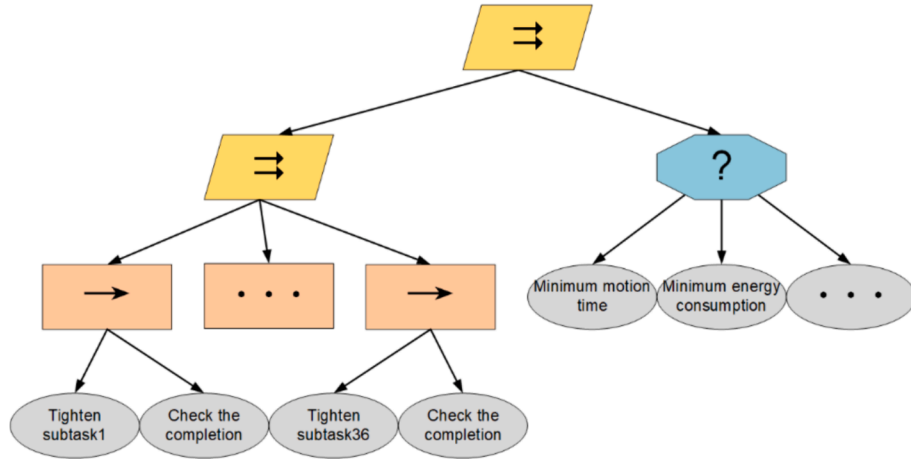


Fig. 6. The cognition model for task agents based on BT.

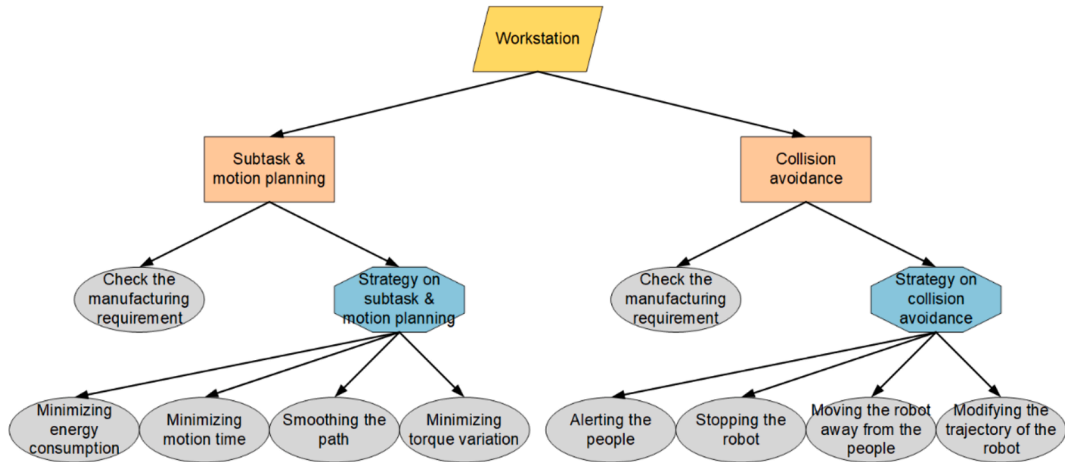


Fig. 7. The cognition model for workstation agents based on BT.

dataset. The final loss function of the DAE is also called the data modality correlation index $Cor_{i,j}$ between data modality i^{th} and data modality j^{th} . The correlation index of each data modality with other data

modalities is calculated by DAE, which in turn constitutes the data modality correlation matrix. When data modality j^{th} are missing in multimodal data, the three largest data modality correlation index

Table 5

Data modalities in screw-tightening and loosening tasks.

Data modalities	Description
1st-modality	The force of the x-axis in the tool center point (TCP) of the robot.
2nd-modality	The force of the y-axis in the TCP of the robot.
3rd-modality	The force of the z-axis in the TCP of the robot.
4th-modality	The torque of rx in TCP of the robot, where rx is a rotation vector representation of the tool orientation.
5th-modality	The torque of ry in TCP of the robot, where ry is a rotation vector representation of the tool orientation.
6th-modality	The torque of rz in TCP of the robot, where rz is a rotation vector representation of the tool orientation.
7th-modality	The torque of rz in TCP of the screwdriver, where rz is a rotation vector representation of the tool orientation.
8th-modality	The force of the z-axis in the TCP of the screwdriver.

$(Cor_{a,j}, Cor_{b,j}, Cor_{c,j})$ and their corresponding DAE models ($DAE_{a,j}$, $DAE_{b,j}$, $DAE_{c,j}$) are selected by querying the data modal correlation matrix. The data modalities (a^{th}, b^{th}, c^{th}) from the multimodal data are input into DAE models ($DAE_{a,j}$, $DAE_{b,j}$, $DAE_{c,j}$) to repair the missing data modality separately. The results of the repair of the missing data modality are j_a^{th} , j_b^{th} , j_c^{th} . The three data modality repair results are then integrated by Eq. (1).

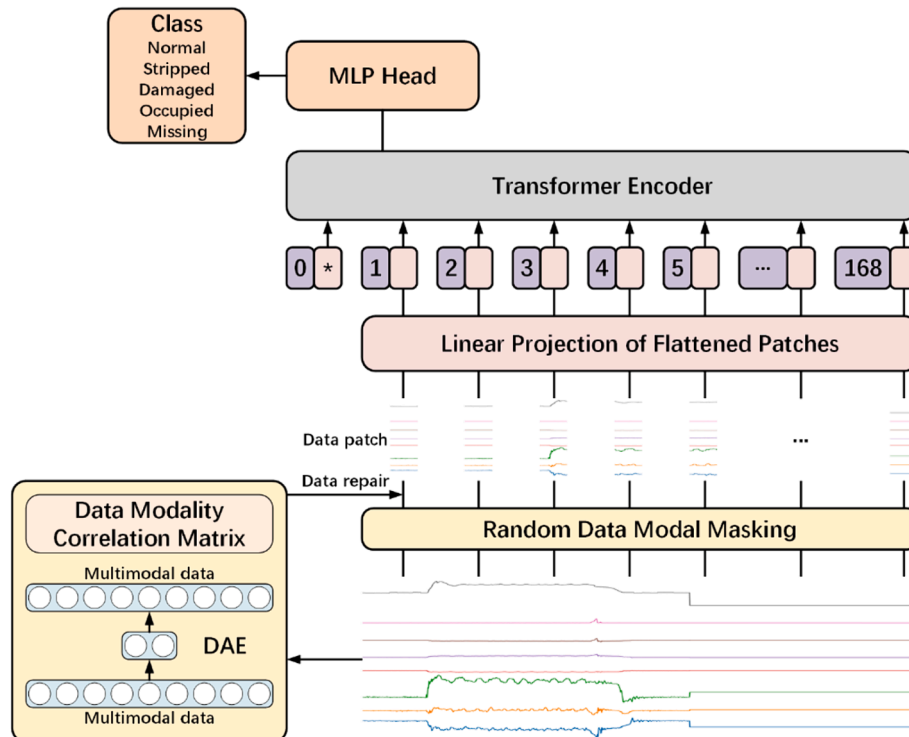
$$j_{repair}^{th} = \frac{e^{Cor_{a,j}}}{\sum_{k=a}^c e^{Cor_{k,j}}} \times j_a^{th} + \frac{e^{Cor_{b,j}}}{\sum_{k=a}^c e^{Cor_{k,j}}} \times j_b^{th} + \frac{e^{Cor_{c,j}}}{\sum_{k=a}^c e^{Cor_{k,j}}} \times j_c^{th} \quad (1)$$

The embedded module is utilized to convert the repaired 8-modality screw-tightening data sample into a serialized data form, which consists of patch embedding and position embedding. Firstly, the embedding module utilizes a convolutional layer to divide the 8-modality data sample into 168 signal patches. Within this convolutional layer, the stride is set to 16, the kernel size to $8*16$, and the number of kernels to 768. Subsequently, the spatial dimensions of this feature map are flattened to generate a vector of size $768*168$. To add categorical and positional information, a $768*1$ category vector and positional encoding

are concatenated with the flattened feature map, yielding a vector of size $768*197$. Following the embedding module, the IMTDAE model proceeds to the Transformer encoder module. Here, the $768*197$ vector derived from the embedding module is input into the Transformer encoder module. The Transformer encoder module comprises 12 Transformer encoder blocks. Each Transformer encoder block is composed of a LayerNorm layer, a Dropout layer, a Multi-Head Attention layer, another LayerNorm layer, a Feed-Forward layer, and a final Dropout layer. Among these layers, the Multi-Head Attention layer assumes a critical role in capturing relationships between different elements of the input vector. The MLP head module constitutes the final module of the IMTDAE model. Within the MLP head module, several components are at play. Firstly, the final output representation from the Transformer encoder module is input into a fully connected layer. Subsequently, a GELU activation function is applied to introduce non-linearity to the output, aiding in modeling complex data relationships. To prevent overfitting, a dropout layer is applied after the GELU activation function, randomly deactivating some neurons during training. Following this, another fully connected layer is employed to further process the output from the previous layer. Ultimately, a final Dropout layer is employed to add regularisation and reduce the risk of overfitting.

4.4.2. Knowledge transfer based on TL

The exploration of the intricate interrelationships between assembly and disassembly tasks have garnered significant attention and research efforts (Ghandi & Masehian, 2015; Torres et al., 2003). While data-driven approaches offer considerable advantages in efficiently and accurately addressing various challenges within the manufacturing industry, their widespread application requires extensive and comprehensive datasets. Comparatively, collecting data for assembly tasks tends to be less challenging and more comprehensive. This is primarily attributable to the higher degree of automation and superior standardization in the assembly phase. Collecting data for the disassembly task is more challenging because of the many uncertainties associated with it. Consequently, if the perception model developed for assembly tasks

**Fig. 8.** The architecture of the IMTDAE.

could be repurposed for disassembly tasks, it presents a promising solution to the challenge of data collection for disassembly tasks. To tackle this issue, particularly in the context of screw-tightening and loosening tasks, this section introduces a knowledge transfer approach based on TL (Pan & Yang, 2010). Specifically, knowledge transfer based on TL uses a small amount of screw-loosening data to fine-tune the previously obtained Transformer model. The TL architecture for screw-loosening tasks is elucidated in Fig. 9. Similar to the data collection and processing detailed in Subsection 4.4.1, the screw-loosening tasks also entail the collection of 8-modality data. Before the padding pre-processing, the order of the 8-modality data needs to be reversed.

In terms of the architecture, it is crucial to emphasize that the architecture designed for screw-loosening tasks mirrors the architecture designed for screw-tightening tasks. This architectural consistency is pivotal, as it ensures a uniform framework for both tasks, thereby facilitating the knowledge transfer process. Specifically, the parameters of the embedding module and the Transformer encoder module are completely identical to those employed in the Transformer for screw-tightening tasks. By maintaining the “frozen” of these modules, they are excluded from the Transformer fine-tuning process for the screw-loosening tasks. This strategy ensures that the knowledge acquired from the screw-tightening tasks remains intact and unaltered as the Transformer adapts to the screw-loosening tasks. The primary focus of the knowledge transfer based on TL is to leverage the limited amount of data of the screw-loosening tasks to fine-tune and optimize exclusively the MLP head module. The MLP head module, being the task-specific component of the Transformer, necessitates adaptation to suit the unique nuances and requirements of the screw-loosening task. By updating the parameters of the MLP head module using a limited amount of data on the screw-loosening task, the Transformer can effectively acquire related knowledge in the screw-loosening tasks.

4.4.3. Autonomous decision-making based on RL

In the screw-loosening tasks, where the completion state of each subtask can influence subsequent subtasks, the subtask sequence planning becomes particularly critical. To tackle this challenge, this paper proposed a subtask sequence planning approach for the screw-loosening tasks based on a deep Q network (DQN) (Mnih et al., 2015). This approach learns subtask sequence planning strategy by enabling the DQN model to interact with the virtual environment. To streamline the problem, the following assumptions are made in this problem:

- The position of the screws is known and fixed.
- The pose of robot TCP does not change during task execution.
- The acceleration and deceleration processes of the robot are not taken into account and the movement time of the robot is proportional to the moving distance.

The subtask sequence planning for the screw-loosening tasks based on DQN can be effectively formulated as a Markov decision process (MDP) as shown in Fig. 10. The MDP is a mathematical framework designed to model the decision-making process within a random environment, characterized by the Markov property. The MDP of subtask sequence planning for the screw-loosening tasks can be represented as a tuple $(S, A, P, \gamma, R, \cdot)$.

The state set S encompasses all possible states that may occur during the screw-loosening tasks. To reduce the model training time and facilitate the presentation of the results, this section reduces the size of the subtasks without changing the complexity of the problem, reducing the 36 subtasks to 9 subtasks. As illustrated in Fig. 11, the state of the screw-loosening tasks can be effectively represented as a square grid. The “START” grid serves as the initial position of the robot. Following the successful completion of a screw-loosening subtask, the UR5e robot proceeds to move to the designated “STORAGE” grid and collaborates with the UR5 robot to deposit the screw. The grids labeled “S1” to “S9” represent 9 pre-determined screw positions. Icons of robots within the grid indicate the current positions of the robot TCP. Green, red, blue, and yellow dots within the grid represent various states. The screw icon within a grid indicates that there is a screw that needs to be loosened. The observation space can be represented by the following equation:

$$S = [\text{Robot}, \text{Subtask}]^T = \begin{bmatrix} 0100000000 \\ 11234000001 \end{bmatrix} \quad (2)$$

The 11 columns in the observation space matrix represent the “START” grid, the “S1” to “S9” grids, and the “STORAGE” grid. The first row of the observation space matrix represents the TCP position of the UR5e, and the second row represents the state of subtasks. The “START” grid and “STORAGE” grid of the second row in the observation space matrix are fixed to 1. The “S1” to “S9” grids of the second row in the observation space matrix use “0”, “1”, “2”, “3” and “4” to represent the loosened state of the screw as “Unloosened”, “Normal”, “Stripped”, “Missing”, and “Damaged” respectively.

The action set A mainly contains actions that determine the next subtask. The action space can be represented by one-hot encoding as the following vector:

$$A = [100000000] \quad (3)$$

$P: S \times A \times S \rightarrow [0, 1]$ is the state transition probability after the robot loosens a screw and is defined as follows:

$$P = \begin{cases} 0.6, & \text{“Normal”} \\ 0.2, & \text{“Stripped”} \\ 0.1, & \text{“Missing”} \\ 0.1, & \text{“Damaged”} \end{cases} \quad (4)$$

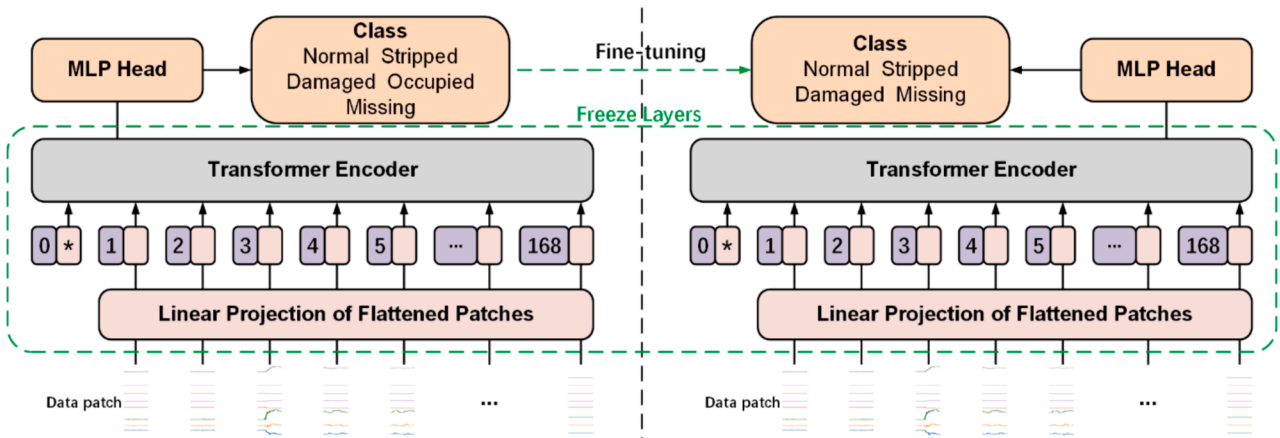


Fig. 9. The TL architecture.

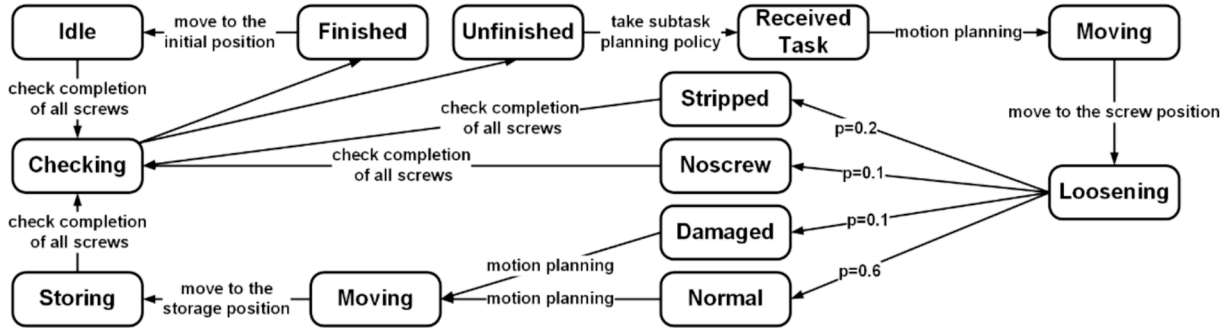


Fig. 10. The MDP for subtask planning.

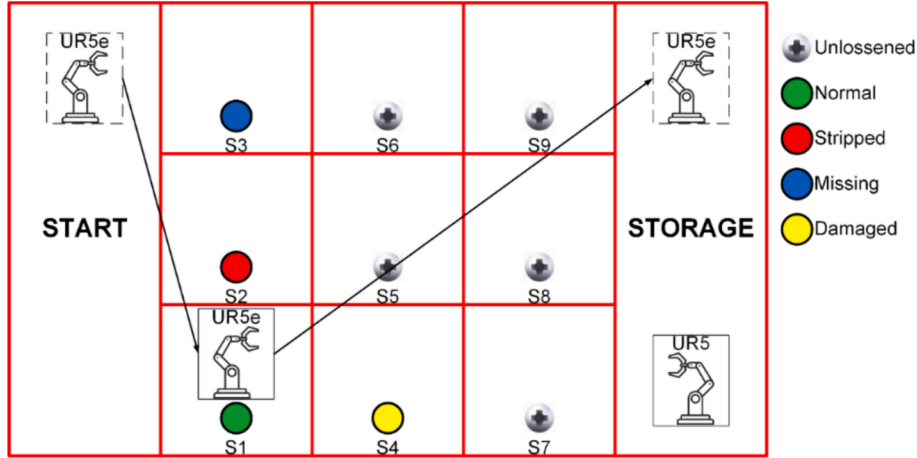


Fig. 11. The state of the screw-loosening tasks.

In contrast, “Missing” and “Damaged” rarely appear in the manufacturing scenario. Therefore, the transition probability for these states is conservatively set at 0.1. Furthermore, the occurrence of the “Stripped” abnormal state is relatively common in screw-loosening tasks. However, given its potential harm to manufacturing equipment, the transition probability associated with this state is set at a cautious 0.2.

γ is the discount factor, which is used to balance the importance of the immediate reward and the future reward and is set to 0.9.

The reward function R is defined as follows. T_{moving} represents the movement time when the robot moves from one grid to another. $R(s, a)$ signifies the reward earned by the robot for executing a screw-loosening subtask. $T_{loosening} = 2$ denotes the fixed screw loosening time. p_{start} and p_{end} represent the start position and target position respectively, and v_{robot} is the speed of the robot. Due to the potential disruption caused by abnormal states, such as “Stripped” and “Damaged”, they extend the screw loosening time to twice the normal loosening time.

$$T_{moving} = \sum \frac{p_{end} - p_{start}}{v_{robot}} \quad (5)$$

$$R(s, a) = \begin{cases} 10 - T_{loosening} - T_{moving} & \text{“Normal”} \\ 10 - T_{loosening} - T_{moving} & \text{“Missing”} \\ 10 - 2 * T_{loosening} - T_{moving} & \text{“Stripped”} \\ 10 - 2 * T_{loosening} - T_{moving} & \text{“Damaged”} \\ 0 - T_{moving} & \text{The grid is Completed} \\ 10000 & \text{Terminated} \end{cases} \quad (6)$$

$$R = \sum R(s, a) \quad (7)$$

The subtask sequence planning strategy $\pi(A|S)$ dictates how the workstation agent should take action in various states. For instance, $\pi(A = “S9”|S = s)$ denote the probability that the next subtask is the “S9” grid when the observation space is s . The value function $V_{\pi}(S)$ provides an essential measure of the expected cumulative reward associated with the subtask sequence planning strategy. These components, the subtask sequence planning strategy π and the value function $V_{\pi}(S)$, form the basis for decision-making and performance evaluation.

Within the DQN, there exists a table that records the expected

reward, denoted as $Q(s, a)$, associated with each action within distinct states. The DQN algorithm selects the best action based on the ϵ -greedy strategy. Then $R(s, a)$ is utilized to assess the effectiveness of the subtask sequence planning strategy and $Q(s, a)$ is updated.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \beta \left(R(s_t, a_t) + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t) \right) \quad (8)$$

Where the parameter β belongs to the range $[0, 1]$, and it plays the role of a learning rate, which governs the speed at which the Q-function is updated in response to a new reward.

Algorithm 1: The DQN algorithm of subtask planning

```

1  Initialise:  $Q(s_t, a_t)$  arbitrarily for all state  $s$  and action  $a$ ;
2  for each episode do
3    Initialise  $s_t$ ;
4    for each step of the episode do
5      Choose a subtask planning action  $a_t$  from  $\mathcal{A}$ 
        using strategy according to  $\epsilon$ -greedy
6      Take action  $a_t$ , observe the next state  $s_{t+1}$  and
        reward  $R(s_t, a_t)$ ;
7       $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \beta \left( R(s_t, a_t) + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t) \right)$ 
8       $s_t \leftarrow s_{t+1}$ 
9      Until  $s_t$  is terminal
10   end
11 end

```

5. Experiments for validation

To demonstrate the feasibility of the proposed CDT framework for MRCMfg and data-driven cognition evolution approaches, this section validates them through case studies. Firstly, the MRCMfg experimental platform and the required dataset are introduced in detail. Then, the result of multimodal perception amongst datasets containing different data modalities is summarized. Subsequently, the effectiveness of knowledge transfer based on TL is verified and the autonomous decision-making strategy for subtask sequence planning in screw-loosening tasks is generated by RL. Finally, a real-time case study is given. All case studies were conducted on a PC equipped with an Intel Core i9-13900k CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 4090 GPU. The numerical results in all case studies are averages obtained after 5 independent tests. It is worth noting that the main purpose of these case studies is to validate the feasibility of the proposed method, and therefore their results may not be optimal.

Table 6
Details of *T&LiR* dataset.

Sub-dataset	Types	Total	Training dataset	Testing dataset
Screw-tightening	Normal	1800	1080	720
	Stripped	540	324	216
	Missing	900	640	360
	Damaged	220	132	88
	Occupied	180	108	72
Screw-loosening	Normal	1800	180	1620
	Stripped	540	54	486
	Missing	900	90	810
	Damaged	220	22	198

5.1. *T&LiR* dataset

In the MRCMfg system for screw-tightening and loosening tasks, DL-based incomplete multimodal perception requires comprehensive training and testing datasets. A dataset named *T&LiR* was collected, which gives a detailed description in Table 6. The 1st to 6th modality data were collected by the internal sensors of the UR5e robot, and the 7th and 8th modality data were collected by the internal sensors of the Onrobot Screwdriver, whose maximum torque is 5 N*m. The sampling frequency is 200 Hz.

T&LiR is a hybrid dataset including two MRCMfg scenarios: screw-

tightening and screw-loosening. The screw type used in this experiment was M6 socket head cap screws. In the “Normal” completion state, the MRCMfg system successfully performed screw-tightening and loosening tasks following formal operating procedures. In the “Missing” completion state, there were no suitable screws for screw-tightening and loosening tasks. To collect data samples in the “Stripped” completion state, the screw heads were polished by tools. “Normal” screws and “Stripped” screws are shown in Fig. 12. For the “Occupied” completion state, small parts were pre-positioned in the screw hole. For the “Damaged” completion state, the screw holes were pre-destroyed with other tools. The “Normal”, “Occupied”, and “Damaged” screw holes are

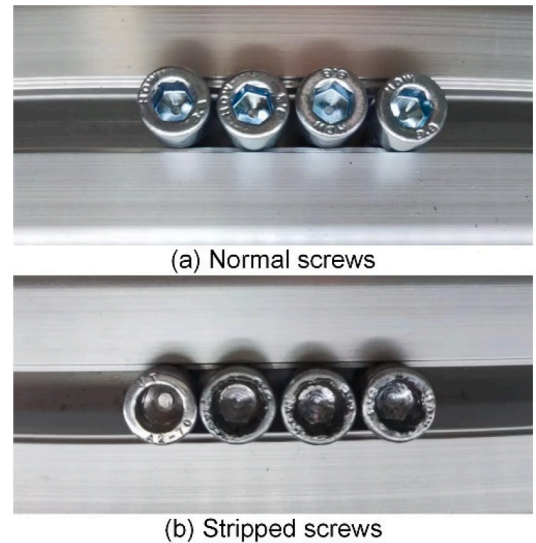


Fig. 12. The “Normal” screws and “Stripped” screws.

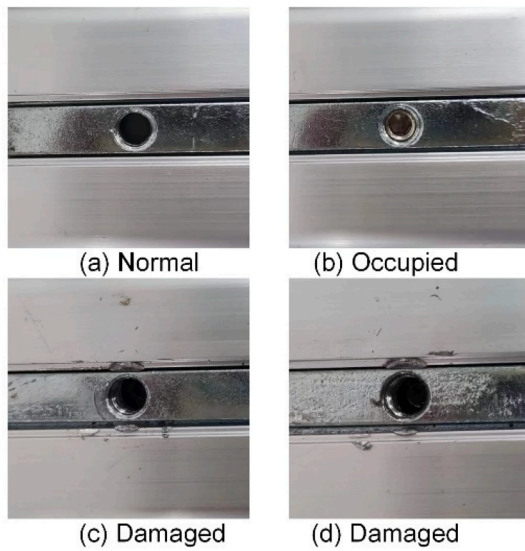


Fig. 13. The abnormal screw holes.

Table 7

The average accuracies and training time of the Transformers for the screw-tightening tasks.

Transformers	Training accuracy (%) $\uparrow$	Testing accuracy (%) $\uparrow$	Training time (s) $\downarrow$
The Transformer using 1st modality data	86.34	86.09	901.57
The Transformer using 2nd modality data	97.09	95.05	912.98
The Transformer using 3rd modality data	94.15	92.01	907.55
The Transformer using 4th modality data	91.10	89.34	903.47
The Transformer using 5th modality data	91.73	91.21	910.87
The Transformer using 6th modality data	88.81	87.94	909.35
The Transformer using 7th modality data	56.29	56.24	904.96
The Transformer using 8th modality data	56.84	60.52	900.40
The Transformer using 8-modality data	100.00	99.85	907.31

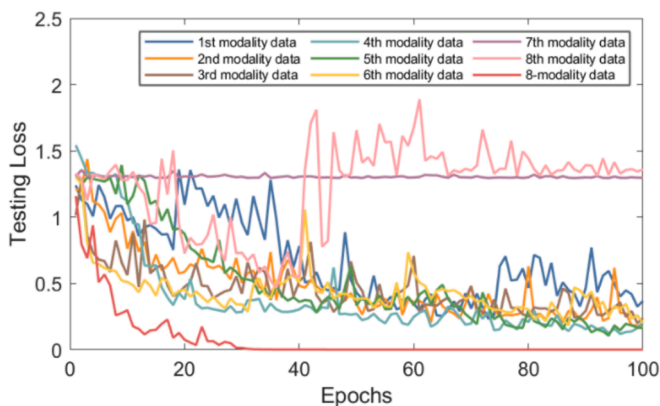


Fig. 14. The testing loss of Transformers for screw-tightening tasks.

illustrated in Fig. 13.

Additionally, the *T&LiR* dataset was divided into a training dataset and a testing dataset. The screw-tightening sub-dataset was randomly

divided into the training dataset and testing dataset in a ratio of 6:4, while the screw-loosening dataset was randomly divided into the training dataset and testing dataset in a ratio of 1:9. The screw-loosening dataset was divided in such a way as to simulate the lack of training data.

5.2. Case study 1: incomplete multimodal perception

Firstly, to validate the multimodal perception performance of the Transformer in the screw-tightening tasks with a sufficient dataset, experiments are conducted on an 8-modality screw-tightening sub-dataset and 8 single-modality screw-tightening sub-dataset. The default optimizer of the Transformer is SGD, with a learning rate of 0.001. Additionally, all the Transformers are trained for 100 epochs with a batch size of 16.

Table 7 shows the results in the screw-tightening sub-dataset. It can be observed that the Transformer using 8-modality data shows better training accuracy and testing accuracy. Different Transformers with different single-modality data show a huge performance gap. The maximum testing accuracy of Transformers using single-modality data is 95.05 %. The difference in training time is very small. This can also prove that the parameter size and training time of the Transformer architecture do not increase dramatically with the increase in data modality. Fig. 14. shows the testing loss of Transformers. It is evident that when training the Transformer with sufficient training data samples, the training loss of the Transformer using 8-modality data exhibits a swift decline and stabilizes after approximately 40 epochs. Conversely, when using a single data modality, the training loss demonstrates notable fluctuations. It indicates that adding data modality in the dataset improves the training performance of the Transformer.

The confusion matrixes of Transformers are shown in Fig. 15. From the confusion matrixes using 7th modality data and 8th modality data, these Transformers predict most of the samples as “Normal”, largely due to the class-imbalance. At the same time, the completion state perception errors of the other 6 Transformers using a single data modality mainly occur in “Damaged” and “Occupied”. However, by strengthening the sample characteristics of the 8-modality data, the Transformer using 8-modality data significantly improves the completion state perception accuracy of all completion states and solves the issue of class-imbalance.

The above experiments have demonstrated the superior performance of the Transformer for multimodal perception. However, when some data modalities are missing, the perception accuracy of the Transformer drops sharply. Table 8 illustrates the performance of the Transformer and IMTDAE in the incomplete multimodal perception condition. Firstly, the performance of the above Transformer is verified using an incomplete multimodal testing dataset that randomly masks 0–2 data modalities. The results show that the Transformer using 8-modality suffers from severe overfitting when tested on the incomplete multimodal testing dataset. Additionally, the results also show that the perception accuracy of the Transformer using incomplete 8-modality data is improved to 97.60 %. However, the Transformer using incomplete 8-modality data still struggles to meet practical needs. Meanwhile, the IMTDAE using incomplete 8-modality data shows superior performance. The IMTDAE has a maximum testing accuracy of 99.26 %. It can also be seen from the confusion matrix shown in Fig. 16. that the IMFDAE has excellent incomplete multimodal perception performance.

5.3. Case study 2: knowledge transfer

In Case Study 2, our objective is to affirm the effectiveness of TL-based knowledge transfer within the screw-loosening tasks, even in the face of a lack of data samples. Experiments are conducted on 8-modality screw-loosening sub-dataset. The training details are similar to Case Study 1. Additionally, to make the Transformer fine-tuning more efficient, cosine learning rate decay is used in knowledge transfer. The cosine learning rate decay factor is set to 0.01.

From the results shown in Table 9, the average testing accuracy of

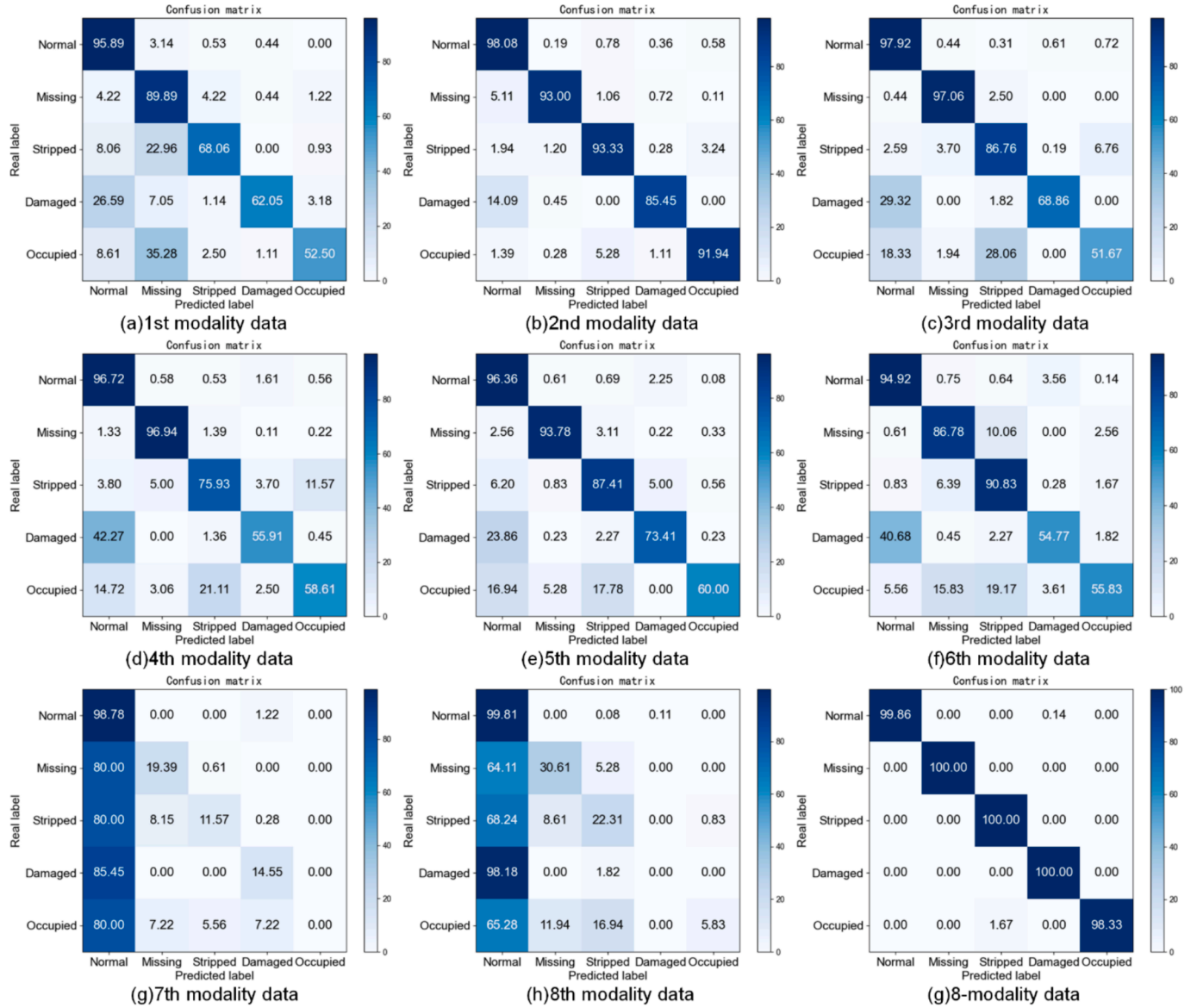


Fig. 15. The confusion matrix of the Transformers for the screw-tightening tasks.

Table 8

Comparisons between Transformer and IMTDAE.

Training data	Training accuracy (%)↑	Testing accuracy (%)↑	Training time (s)↓
The Transformer using 8-modality data	100.00	85.15	893.04
The Transformer using incomplete 8-modality data	98.78	97.60	893.72
The IMTDAE model using incomplete 8-modality data	99.50	99.26	1012.50

the Transformer for the screw-loosening tasks can only reach 97.60 %. TL fine-tunes the Transformer for the screw-tightening tasks with a limited screw-loosening training sub-dataset, which in turn leads to the knowledge transfer of the Transformer from the screw-tightening tasks to the screw-loosening tasks. The average testing accuracy of the Transformer with TL is 99.74 %, which is much higher than the average testing accuracy of the Transformer without TL. The results show that the TL can effectively improve the training accuracy and testing accuracy of the Transformer with a limited training dataset. In addition, the

average training time of the Transformer with TL for the screw-loosening tasks is 414.72 s, which is less than the average training time of the Transformer without TL. The results show that the TL not only improves the accuracy of the Transformer for the screw-loosening tasks but also effectively reduces the training time.

Based on the findings illustrated in Fig. 17, when utilizing TL to fine-tune the previously obtained Transformer to train the Transformer for the screw-loosening tasks, the completion state perception performance is significantly improved. In this scenario, the training loss of the Transformer with TL experiences a swift decline and reaches stability after approximately 60 epochs. This swift decline in the training loss is attributable to the fact that the TL leverages the knowledge acquired from the screw-tightening tasks to the screw-loosening tasks. This knowledge transfer makes the process of training the Transformer for screw-loosening tasks faster and more efficient.

Fig. 18 shows the confusion matrixes of the Transformer with and without TL. The issue of class-imbalance remains unresolved in cases, even using multiple data modalities to enhance completion state perception performance. However, knowledge transfer based on TL can draw on knowledge obtained from the screw-tightening sub-dataset to improve the perception accuracy of “Damaged”.

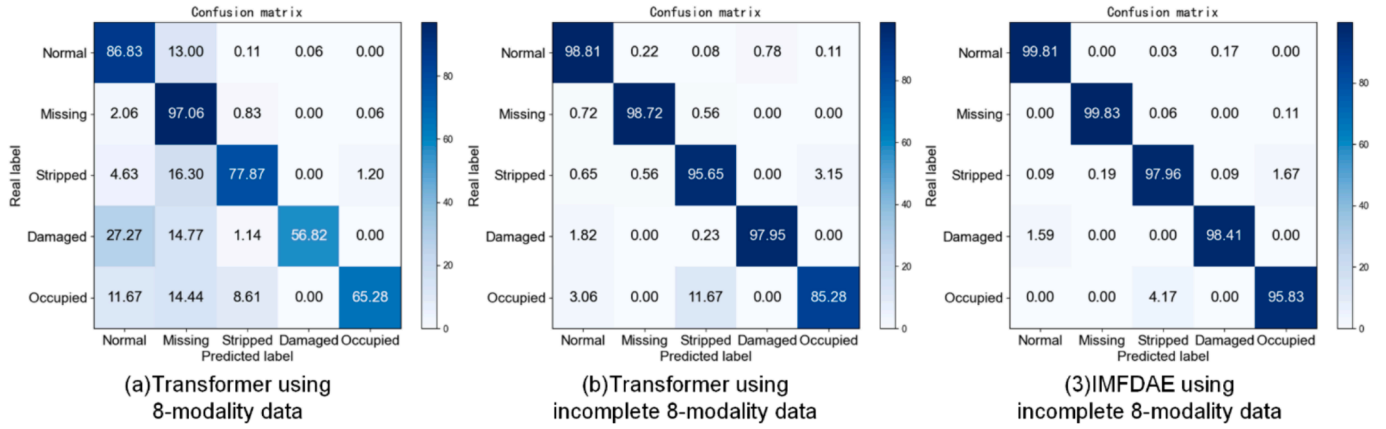


Fig. 16. The confusion matrix of the Transformer and IMFDAE.

Table 9

The average accuracies and training time of the Transformer for the screw-loosening tasks.

Models	Training accuracy (%) [†]	Testing accuracy (%) [†]	Training time (s) [‡]
The Transformer model without TL	98.96	97.60	496.93
The Transformer model with TL	100.00	99.74	414.72

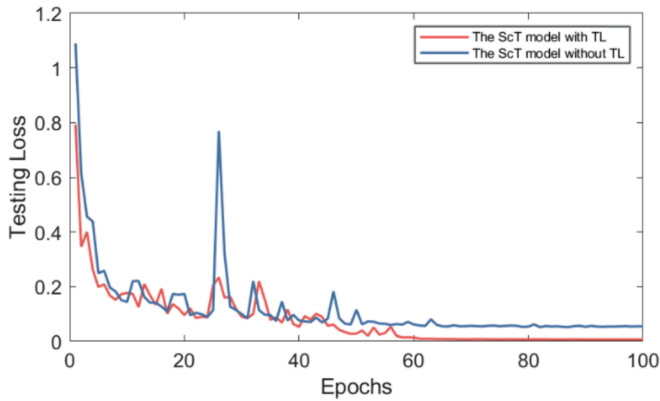


Fig. 17. The testing loss of the Transformer for the screw-loosening tasks.

5.4. Case study 3: autonomous decision-making

To verify the effectiveness of the DQN for autonomous decision-making and subtask sequence planning in complex and uncertain screw-loosening tasks, this section builds a DQN training environment for screw-loosening subtask planning as shown in Fig. 11. In this training environment, the DQN needs to decide on the next subtask based on the current robot location and the completion states of each screw-loosening subtask. The training reward and training loss of the DQN for the screw-loosening tasks is shown in Figs. 19 and 20, respectively. It is obvious from the figure that with the steps increase, the training reward and training loss of DQN gradually converge at the same time.

For the MRCMfg system without the cognition capabilities, it can only complete 9 subtasks from 1 to 9 sequentially. As shown in Table 10, the average total time to complete all 9 subtasks is 91.35 s. However, the MRCMfg system with cognition capability can make real-time decisions on the optimal subtasks sequence based on the state of the manufacturing task. In the context of the screw-loosening tasks, achieving the shortest task completion time primarily hinges on

minimizing the total movement time of robots. In contrast, the MRCMfg system with cognition capabilities takes 89.86 s to complete the same 9 screw-loosening subtasks through real-time subtask sequence planning.

Although the number of screw-loosening subtasks included in this case study is small, the results show that autonomous decision-making based on RL can effectively improve the efficiency of the MRCMfg system. In this case study, compared with the pre-planning subtask sequence, the completion time of the subtask sequence obtained by DQN-based autonomous decision can be reduced by about 1.5 s. Nevertheless, the method still shows great application potential. Firstly, with the enhancement of social environmental awareness, the scale of the disassembly tasks will continue to increase. The effectiveness of autonomous decision-making in MRCMfg will be more significantly reflected in large-scale disassembly tasks. In addition, the task scenario of this case study is relatively simple. The complex manufacturing tasks require the MRCMfg system to adjust and make decisions in real-time according to the manufacturing environment. The effectiveness of autonomous decision-making for MRCMfg will be more obvious when the manufacturing task is more complex and uncertain. Also, this will require more time and computational resources to train the DQN model.

5.5. Real-time case study for the screw-loosening tasks on MRCMfg

For the real-time case study, the real MRCMfg system is used to execute the screw-loosening tasks shown in Fig. 21. The action nodes of cognition models for robot agents are generated by URScript to control the motion of robots. Meanwhile, the state nodes of cognition models for the UR5e robot agent are constructed from the Transformer with TL for the screw-loosening tasks. The subtask planning strategy in the cognition model for the workstation agent is generated in case study 3. This real-time case study is close to the actual manufacturing environment. Fig. 22. shows the snapshots of the MRCMfg system in the real-time case study. In Fig. 22, the target screw is indicated by a red circle, and the MRCMfg system collaboratively loosens a screw that can be divided into the following 6 steps. Since different devices in the MRCMfg system have different time delays and built-in sampling frequencies, it is necessary to perform linear interpolation and resampling of the acquired 8-modality data before inputting the Transformer with TL for the screw-loosening tasks. Throughout the process, every motion of the real robot must be examined in the virtual models of DT before execution. The upper right corner of each step in Fig. 22 shows the simulation process in the virtual models.

Table 11 shows the results of state perception for screw-loosening tasks in the real-time case study. The real-time case study is repeated 5 times. Since the *T&LiR* dataset was collected in the same scenario as the real-time case study, the Transformer trained on the *T&LiR* dataset has good performance in the real-time case. The perception accuracy of

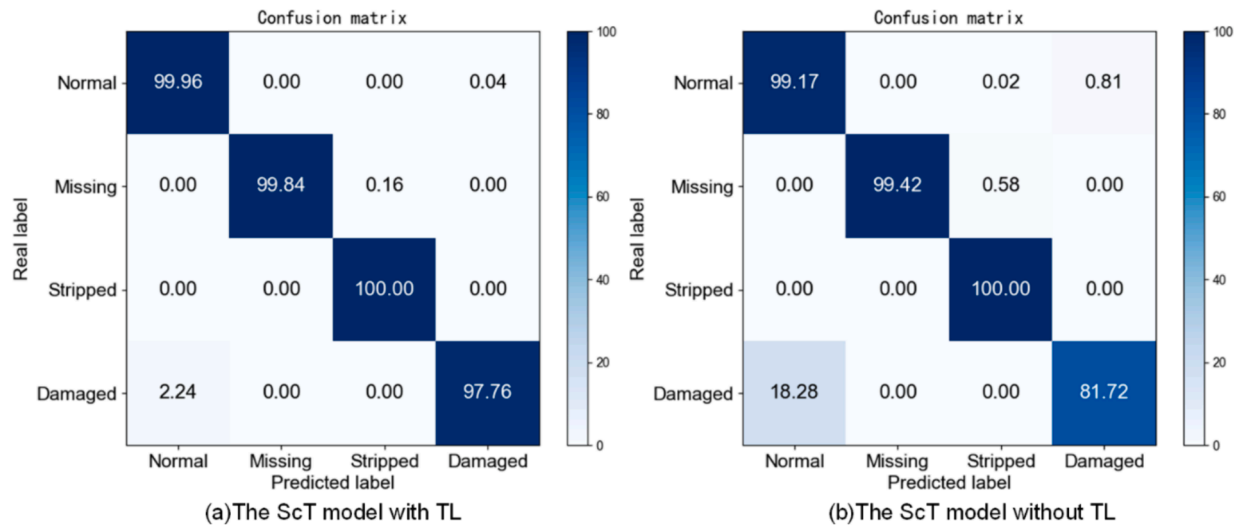


Fig. 18. The confusion matrix of the Transformer for the screw-loosening task.

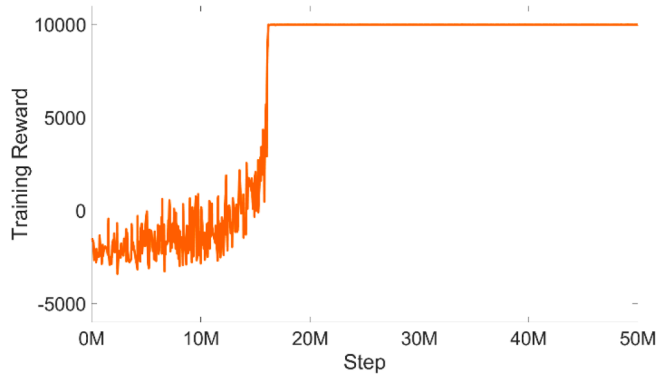


Fig. 19. The training reward for DQN.

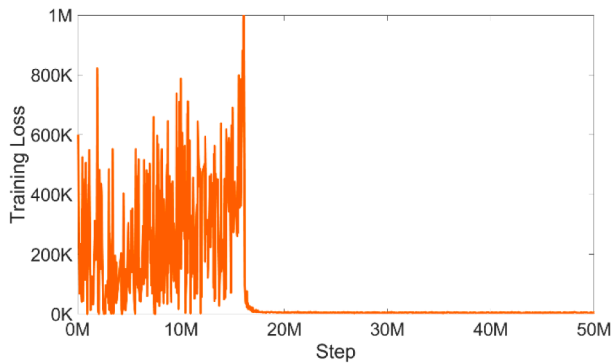


Fig. 20. The training loss for DQN.



Fig. 21. The real screw-loosening tasks.

the real-time case study was 100 % in “Normal”, “Missing” and “Stripped”. However, in “Damage”, a perception error occurs. It must be noted that the perception error counted here can be caused by a variety of reasons. One possible reason is class-imbalance. Additionally, variations in geometric characterization due to repeated experiments on the same “Damaged” screw hole may be another reason.

In case study 3, we make some assumptions to simplify the manufacturing task and manufacturing environment. In the real-time case study, the actions of the MRCMfg system are difficult to represent accurately through an MDP. Therefore, whether RL-based manufacturing task planning strategies can be directly applied to real manufacturing systems has been a hot topic. In Fig. 23, the subtask sequence and time of the MRCMfg system in a real-time case study are shown. Compared to case study 3, the MRCMfg system takes approximately 4.5 times longer to accomplish the same task in the real-time case study than in the MDP model. This reflects that it is difficult to accurately reflect the reality of an MRCMfg system through mathematical models and logic models alone. However, it is fortunate that the RL-based subtask sequence planning strategy does play a role in this real-time case study of the MRCMfg system. The time required to complete the 9 screws loosening subtask is reduced by about 6s. Although the effect of subtask sequence planning is not very obvious, the results prove that the autonomous decision-making approach of MRCMfg based on RL

Table 10

Comparisons between the pre-planning subtask sequence and autonomous decision-making subtask sequence.

Methods	Completion time (s)↓
Pre-planning	91.35
Autonomous decision-making with DQN	89.96

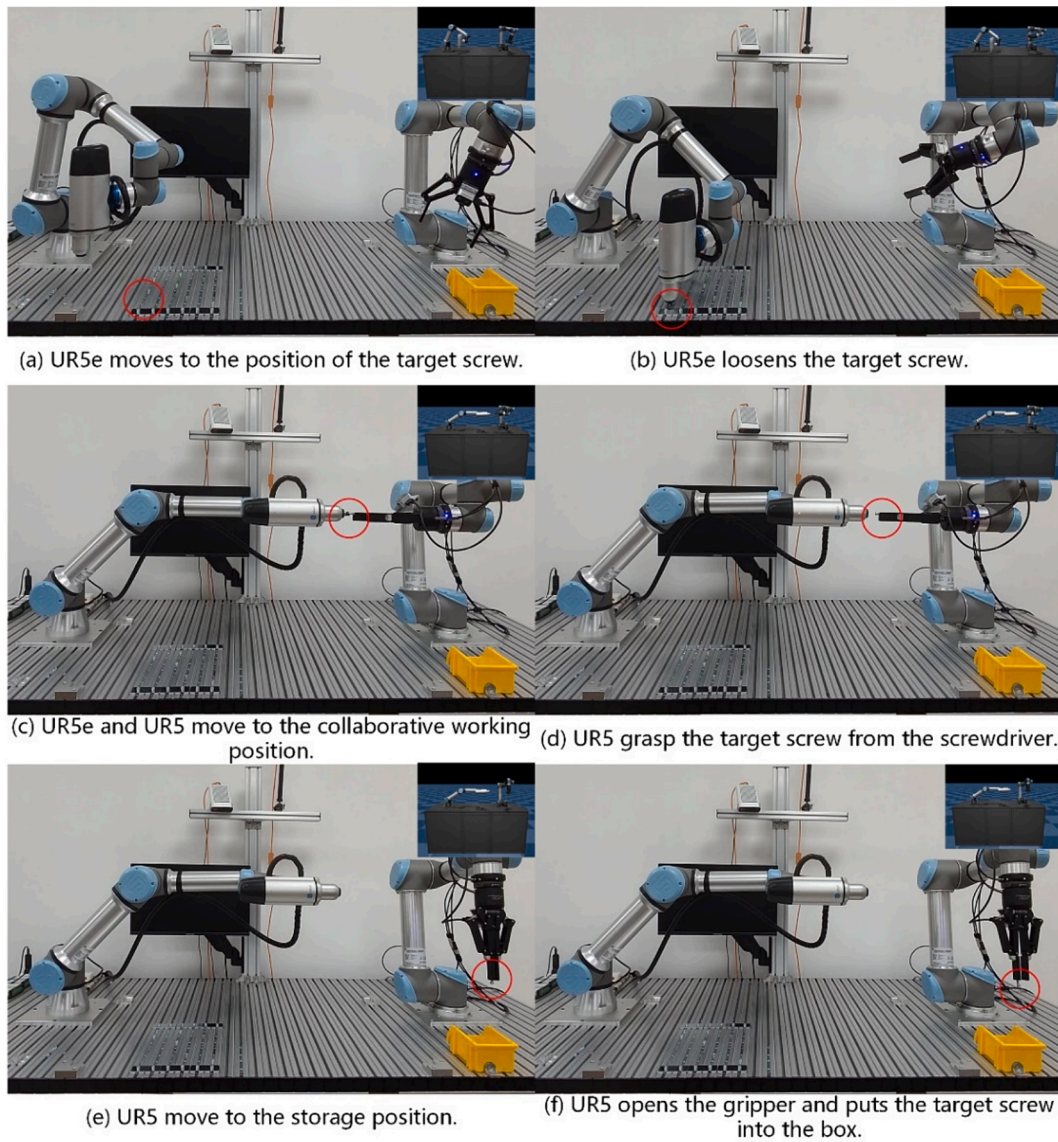


Fig. 22. Snapshots of the MRCMfg system to loosen a screw.

Table 11

Completion state perception in the real-time case study.

Completion	Total	Perception errors ↓	Perception accuracy (%)↓
“Normal”	20	0	100
“Missing”	5	0	100
“Stripped”	15	0	100
“Damaged”	5	1	80

can optimize the MRCMfg process.

6. Conclusion, discussion, and future work

In this paper, a novel CDT framework for MRCMfg is developed to empower MRCMfg with intelligence and cognition capabilities. The innovative characteristics of the CDT framework for MRCMfg include:

- This paper presents a comprehensive CDT framework that strategically combines DT, knowledge-driven approaches, and data-driven approaches. This integration harnesses the individual strengths of these techniques to enhance the intelligence and cognition capabilities of MRCMfg.

- A unified approach for constructing cognition models based on BT is introduced to accommodate the distinctive characteristics of various manufacturing elements. This approach is specifically designed to facilitate the seamless integration of data-driven approaches and knowledge-driven approaches.
- The cognition evolution of MRCMfg based on multiple data-driven approaches is proposed to adapt to the dynamic manufacturing environment, including multimodal perception, knowledge transfer, and autonomous decision-making. Among them, an IMTDAE approach is proposed to realize incomplete multimodal perception.
- The proposed cognition evolution approaches are validated by four case studies with different goals and complexities. The results demonstrate the effectiveness and feasibility of the proposed CDT framework.

In the real manufacturing process, the CDT framework can be widely used to support complex and uncertain manufacturing tasks and manufacturing systems, such as automotive transmission assembly and disassembly based on MRCMfg.

In addition, we would like to discuss a few interesting issues in the experiment. The first issue is the validity of screwdriver torque data for the completion state perception. In some studies, screwdriver torque

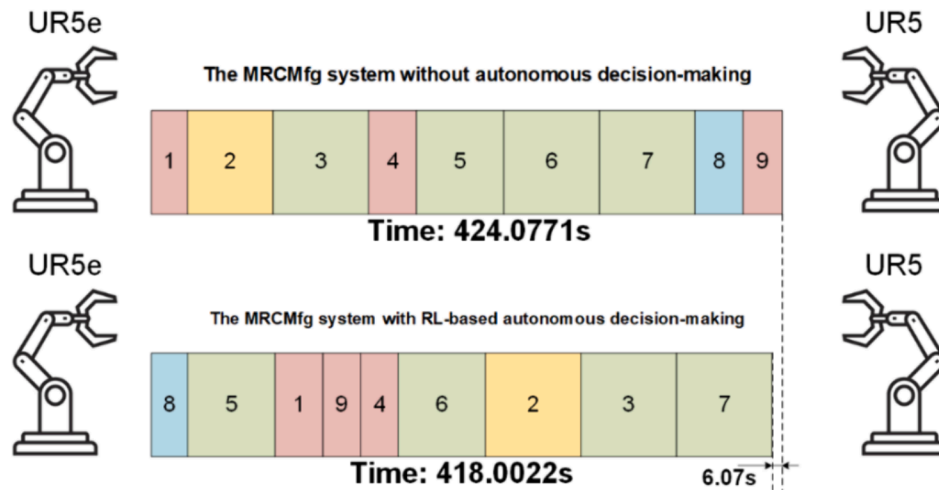


Fig. 23. The subtask sequence and time in the real-time case study.

data is considered to be the best completion state perception data (Jia et al., 2019). In this paper, however, the screwdriver torque data works very poorly. This result can be caused by several reasons. Firstly, the screwdriver we used is oriented to scientific research, and its data collection accuracy may not meet industrial requirements. Then, the severe class-imbalance issue and the structure of the IMTDAE model may also constrain the usefulness of this data type. The second issue is the relationship between assembly and disassembly tasks in manufacturing processes. During the manual manufacturing stage, experience and knowledge of the assembly process are often used to guide the disassembly process. In this paper, we demonstrate again that the experience and knowledge of the assembly process can be used equally well to guide the disassembly process in RMfg.

There are some further research issues to be explored. For instance, the task planning and motion planning of MRCMfg is still not efficient enough to support a collaborative plan. In addition, the MRCMfg system in the case study can successfully perform the task on the assumption that all three cognition models could be efficiently generated. Once the accuracy of any cognition model decreases drastically, such as the incomplete multimodal perception mentioned in case study 1, it will seriously affect the operation of the other two cognitive models.

CRediT authorship contribution statement

Wenjun Xu: Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Hang Yang:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Zhenrui Ji:** Writing – review & editing, Visualization. **Mengyuan Ba:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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